

2.4 Intermediate bonding and bond polarity

Students will be assessed on their ability to:

- a explain the meaning of the term *electronegativity* as applied to atoms in a covalent bond
- b recall that ionic and covalent bonding are the extremes of a continuum of bonding type and explain this in terms of electronegativity differences leading to bond polarity in bonds and molecules, and to ionic bonding if the electronegativity is large enough
- c distinguish between polar bonds and polar molecules and be able to predict whether or not a given molecule is likely to be polar
- d carry out experiments to determine the effect of an electrostatic force on jets of liquids and use the results to determine whether the molecules are polar or non-polar.